

M003 Week 4 — English Worksheet

Topic: The Ground Plane (x, z) · **Course:** 3D Coordinates · **Time:** about 45 minutes

For three weeks you drew standing up on a **wall** with (x, y). This week you look **down** at the **floor**, where the two numbers are **(x, z)**: x is how far **across**, z is how far **forward**. A nested loop fills a whole floor — x runs across each row, z steps forward to the next row.

1 · Predict 🧠

Read each set of steps. Before you imagine it happening, write what you think you will see.

```
place gold block at (3, 0)
place gold block at (3, 5)
```

On the floor these are (x, z). The two blocks share the same x. Are they close together or far apart? Which number changed, and what does it mean?


```
set z to 0
repeat 5 times:
  place stone block at (2, z)
  add 1 to z
```

Which way does this line run — across or forward? How long is it? Which number stays the same?


```
set z to 0
repeat 3 times:
  set x to 0
  repeat 4 times:
    place stone block at (x, z)
    add 1 to x
  add 1 to z
```

The inner loop runs x across, the outer loop steps z forward. What shape covers the floor — a line or a rectangle? How wide and how deep?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The path should run **forward** at $x = 2$, from $(2, 0)$ to $(2, 4)$. x should stay 2. Right now it runs across instead.

```
set n to 0
repeat 5 times:
  place stone block at (n, 2)
  add 1 to n
```

Hint: look at which slot n is in — the x slot (across) or the z slot (forward).

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should fill a floor **5 across and 3 forward**. Right now each row lands on top of the last one, so you get a thin line instead of a rectangle.

```
set z to 0
repeat 3 times:
  set x to 0
  repeat 5 times:
    place stone block at (x, z)
    add 1 to x
```

Hint: after each row, the floor needs to step forward to the next z .

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The floor block should land at **(4, 6)** — 4 across, 6 forward. Right now it lands somewhere else.

```
place gold block at (6, 4)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Fill a **floor pad** with a nested loop, then lay a **path** across it that runs forward (z grows) or across (x grows). When you finish, come back here.

Send a photo or video of your floor, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

My floor pad is ... across and ... forward.

The inner loop grew the number ... and the outer loop grew the number ...

My path runs ... because it grows the number ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you look down at your floor. Talk through it like you are teaching someone who only knew the wall before. Try to use these words: **floor, x, z, forward, across**.

1. Show the block at (0, 0) where the floor begins.
2. Show your code and say which loop runs **across** and which steps **forward**.
3. Walk along your path and say which number is growing as you move.
4. Say in your own words why the floor uses **x and z**, not x and y.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.